

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

Data Preview

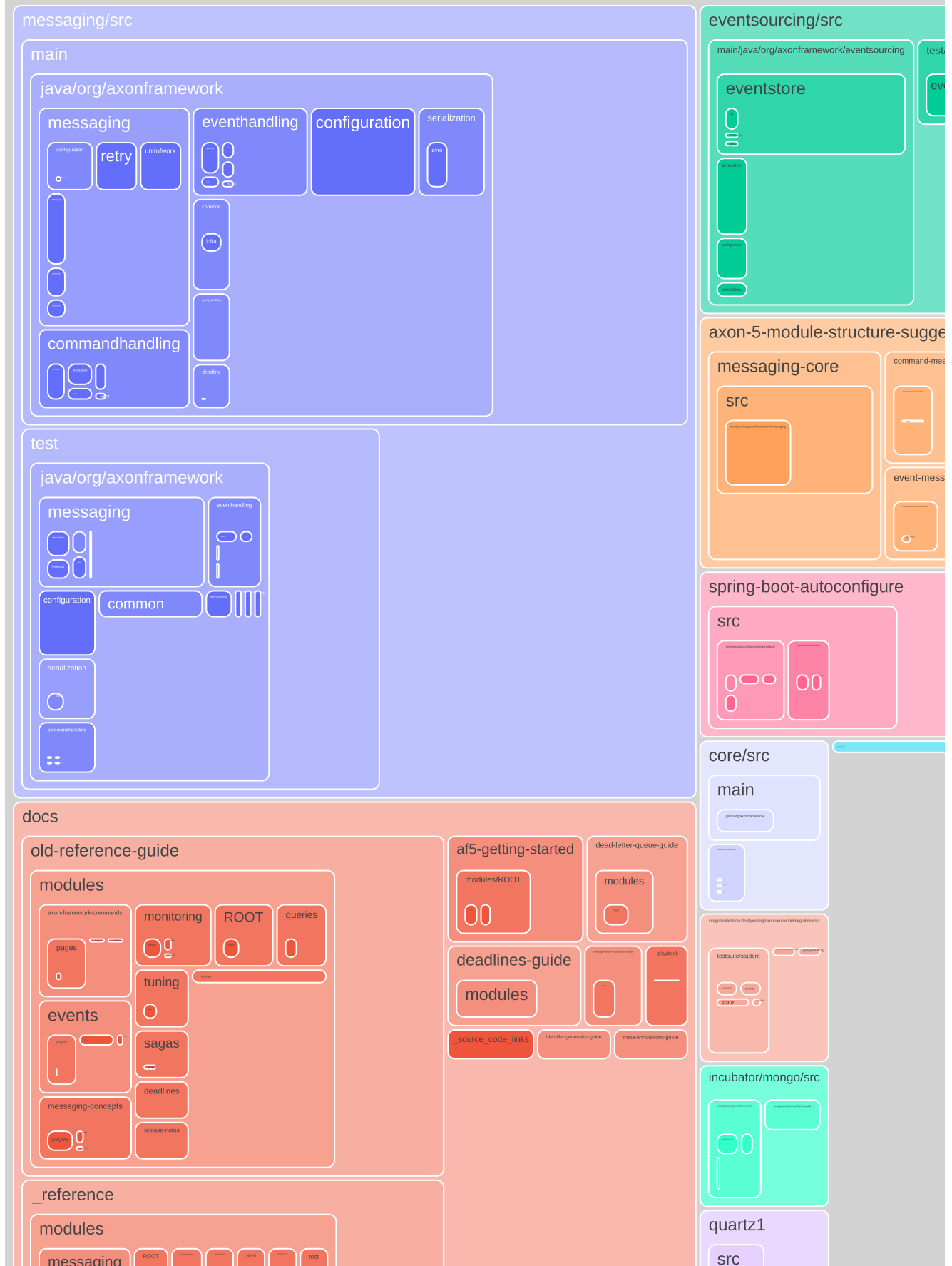
	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	316.000000	316.000000	316.000000	316.000000	316.000000	316.000000
mean	14.145570	8.588608	404.939873	396.218354	630.648734	562.832278
std	58.449905	9.187382	1776.388585	1038.459317	1211.536383	1220.149536
min	1.000000	2.000000	2.000000	1.000000	3.000000	0.000000
25%	1.000000	3.000000	24.000000	3.000000	47.000000	3.000000
50%	4.000000	7.000000	72.500000	4.000000	197.000000	116.000000
75%	9.000000	9.000000	189.750000	262.000000	315.000000	304.000000
max	936.000000	90.000000	27302.000000	4803.000000	4891.000000	4802.000000

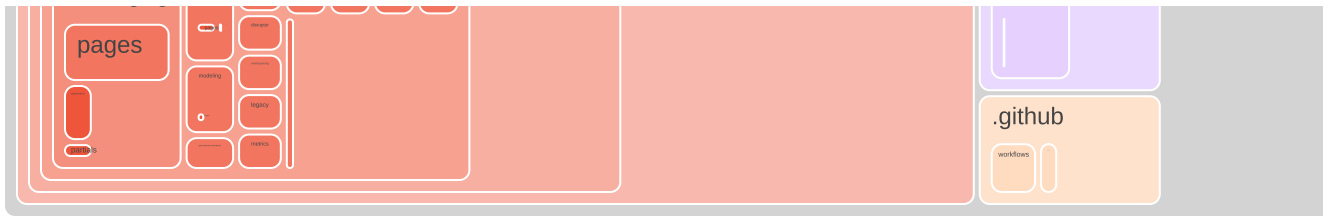
	directoryPath	directoryParentPath
0	AxonFramework-4.11.1/idea/copyright	AxonFramework-4.11.1/idea
1	AxonFramework-4.11.1/idea/inspectionProfiles	AxonFramework-4.11.1/idea
2	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src/test/java/org/axonframework/messaging	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src test/java/org/
3	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/command	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector
4	AxonFramework-4.11.1/core/src/test/java/org/axonframework/commandhandling/disruptor	AxonFramework-4.11.1/core/src/test/java/org/axonframework
5	AxonFramework-4.11.1/core/src/test/java/org/axonframework/contextsupport/spring	AxonFramework-4.11.1/core/src/test/java/org/axonframework
6	AxonFramework-4.11.1/core/src/test/java/org/axonframework/eventhandling/scheduling/quartz	AxonFramework-4.11.1/core/src/test/java/org/axonframework eventl
7	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbook
8	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROOT
9	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.1/docs/_reference/modules/ROOT
10	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
12	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuration
13	AxonFramework-4.11.1/docs/_reference/modules/configuration/partials	AxonFramework-4.11.1/docs/_reference/modules/configuration
14	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
15	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.1/docs/_reference/modules/disruptor
16	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
17	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
18	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legacy
19	AxonFramework-4.11.1/docs/_reference/modules/legacy/partials	AxonFramework-4.11.1/docs/_reference/modules/legacy
20	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metrics
21	AxonFramework-4.11.1/docs/_reference/modules/metrics/partials	AxonFramework-4.11.1/docs/_reference/modules/metrics
22	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micrometer
23	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.1/docs/_reference/modules/micrometer
24	AxonFramework-4.11.1/docs/_reference/modules/modeling/partials	AxonFramework-4.11.1/docs/_reference/modules/modeling
25	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/spring
26	AxonFramework-4.11.1/docs/_reference/modules/spring/partials	AxonFramework-4.11.1/docs/_reference/modules/spring
27	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
28	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
29	AxonFramework-4.11.1/docs/_reference/modules/test/pages	AxonFramework-4.11.1/docs/_reference/modules/test

Number of files per directory

Directories and their file count

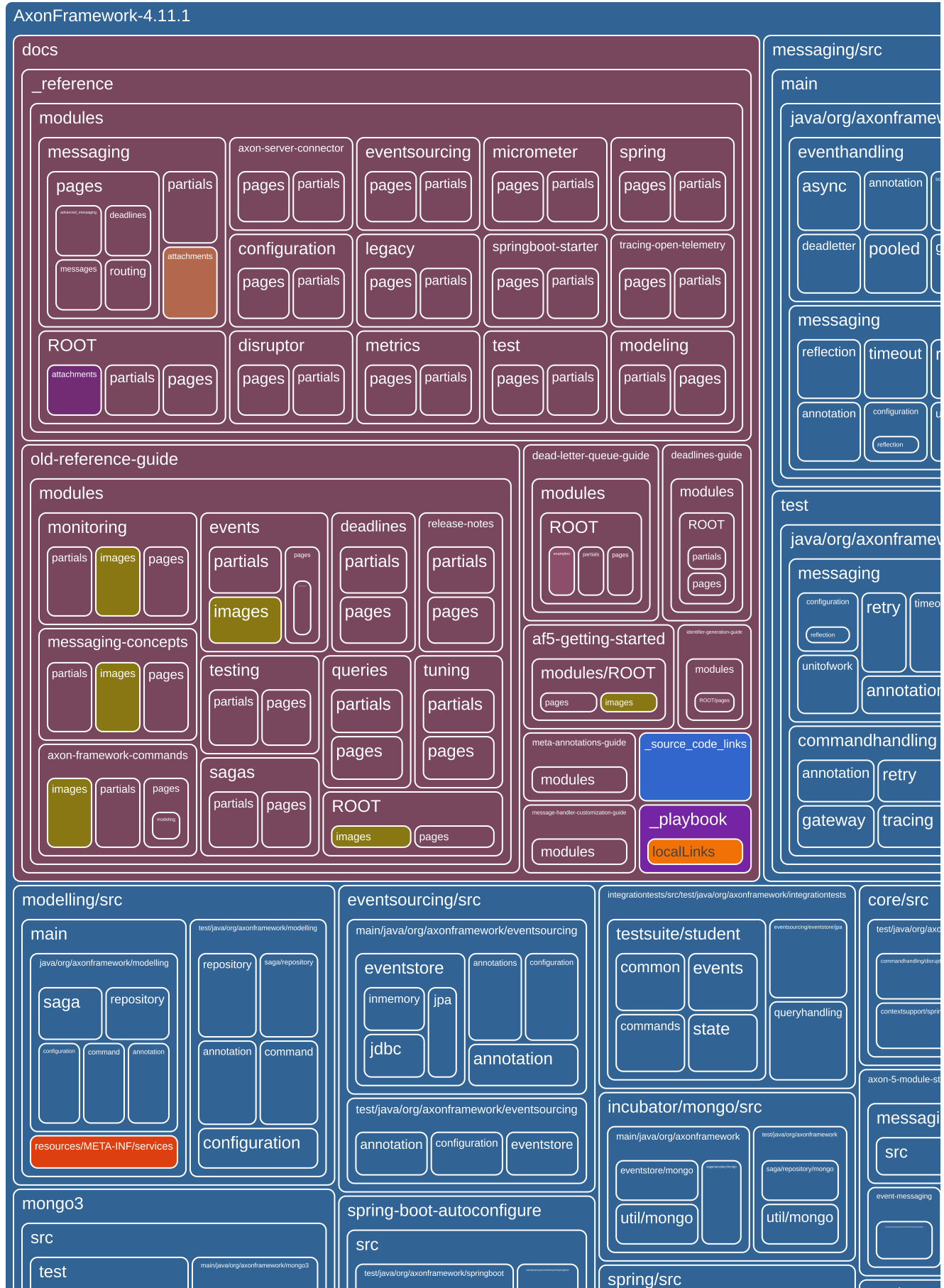
AxonFramework-4.11.1

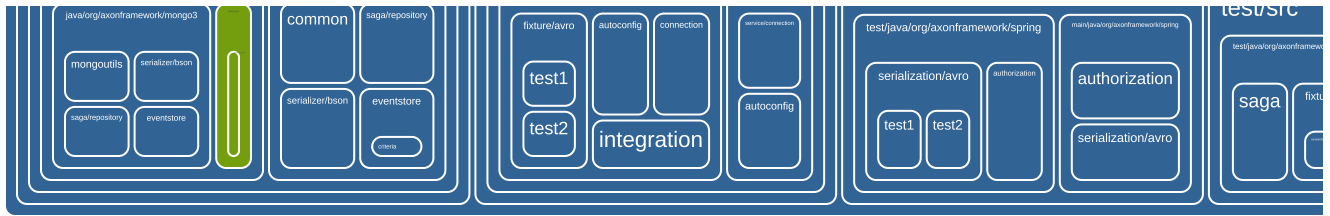




Most frequent file extension per directory

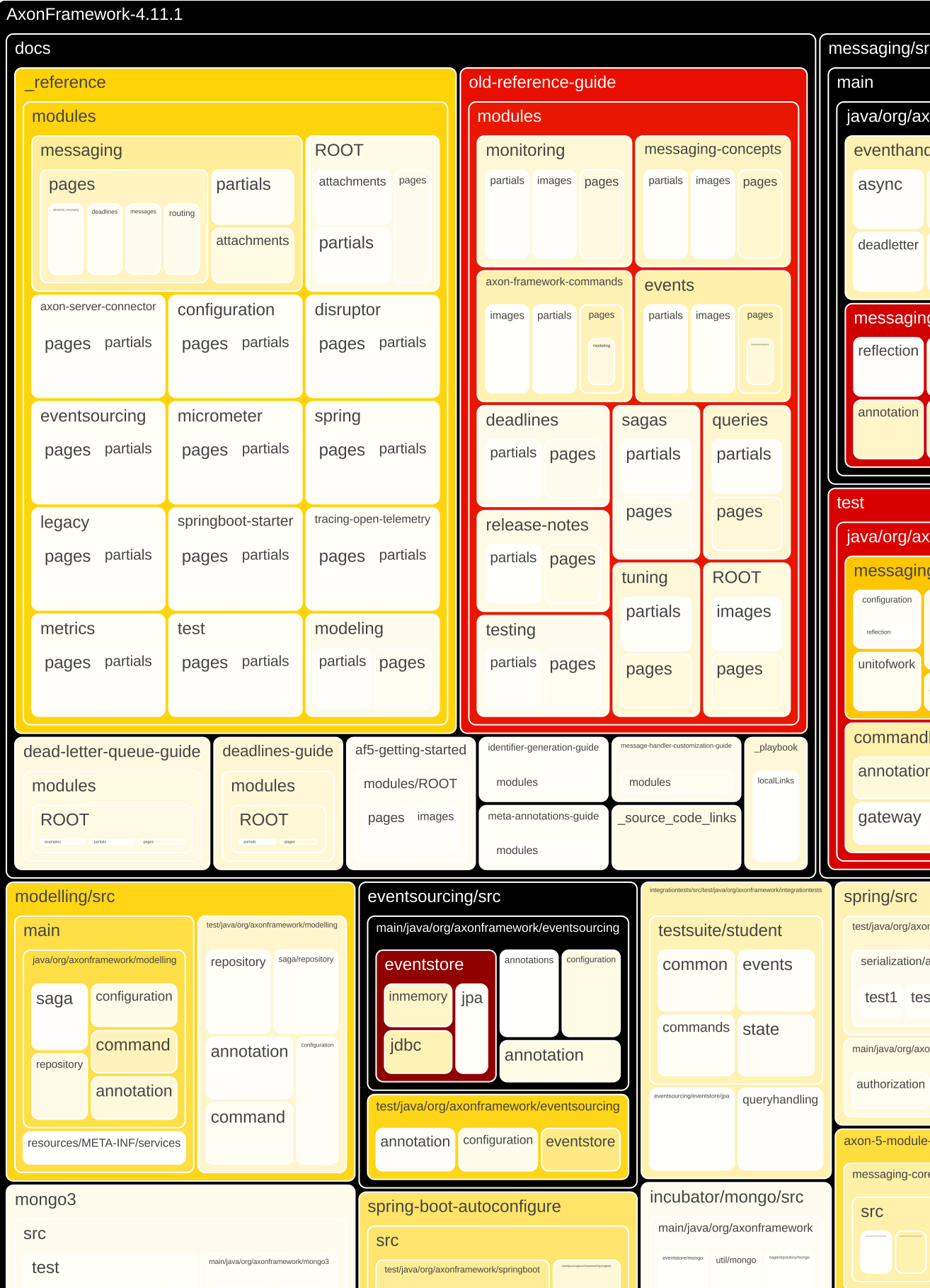
Most frequent file extension per directory

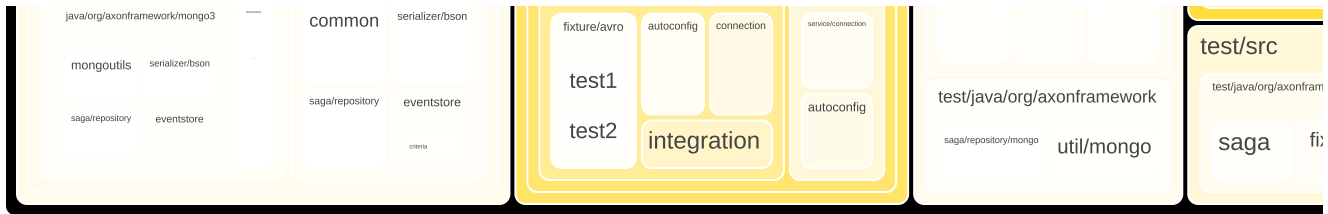




Number of commits per directory

Number of git commits

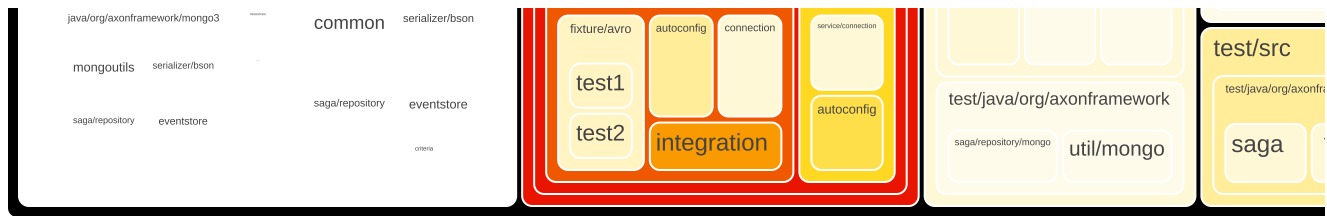




Number of distinct authors per directory

Number of distinct commit authors

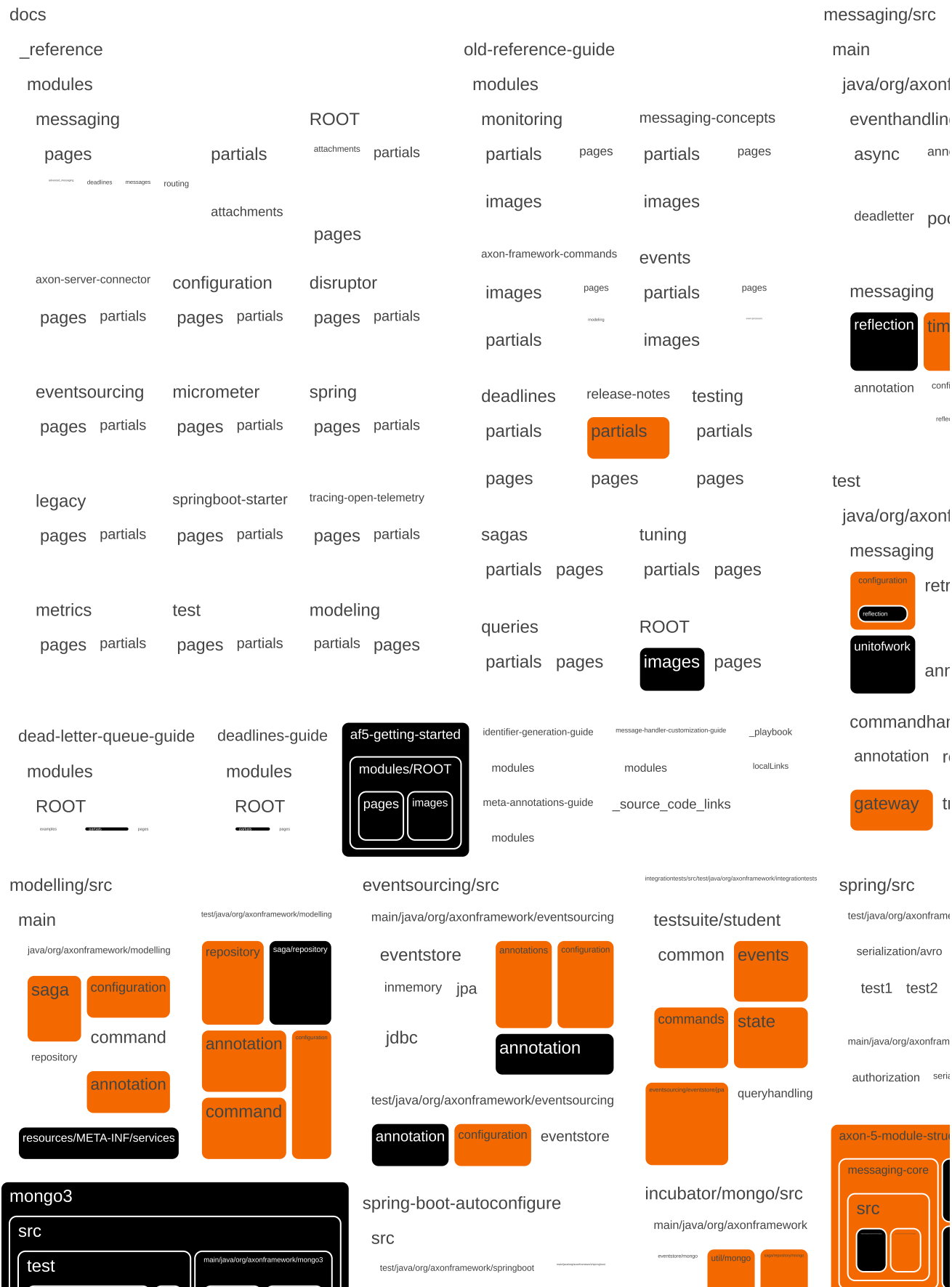


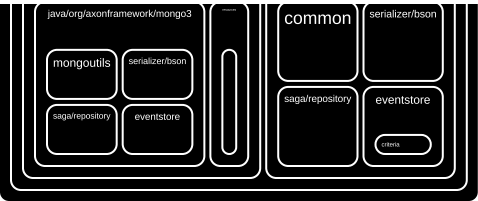


Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.1





fixture/avro autoconfig connection service/connection

test1

test2 integration

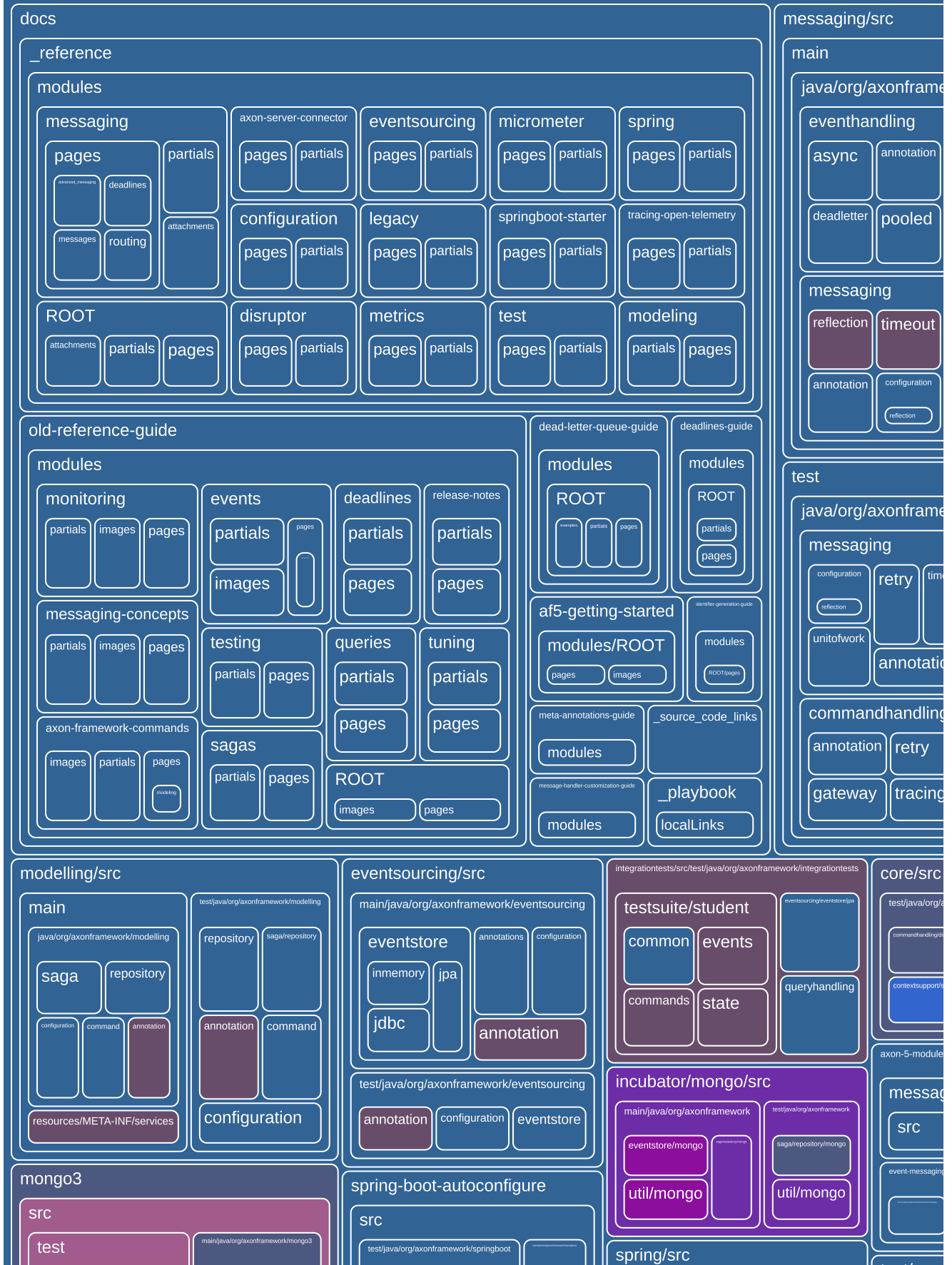
autoconfig

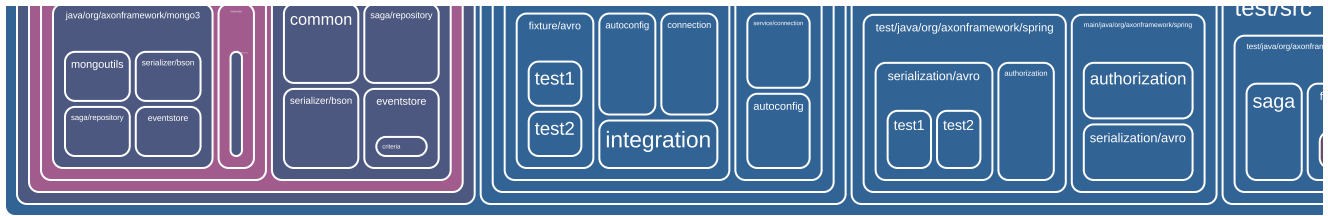


Main author per directory

Main authors with highest number of commits

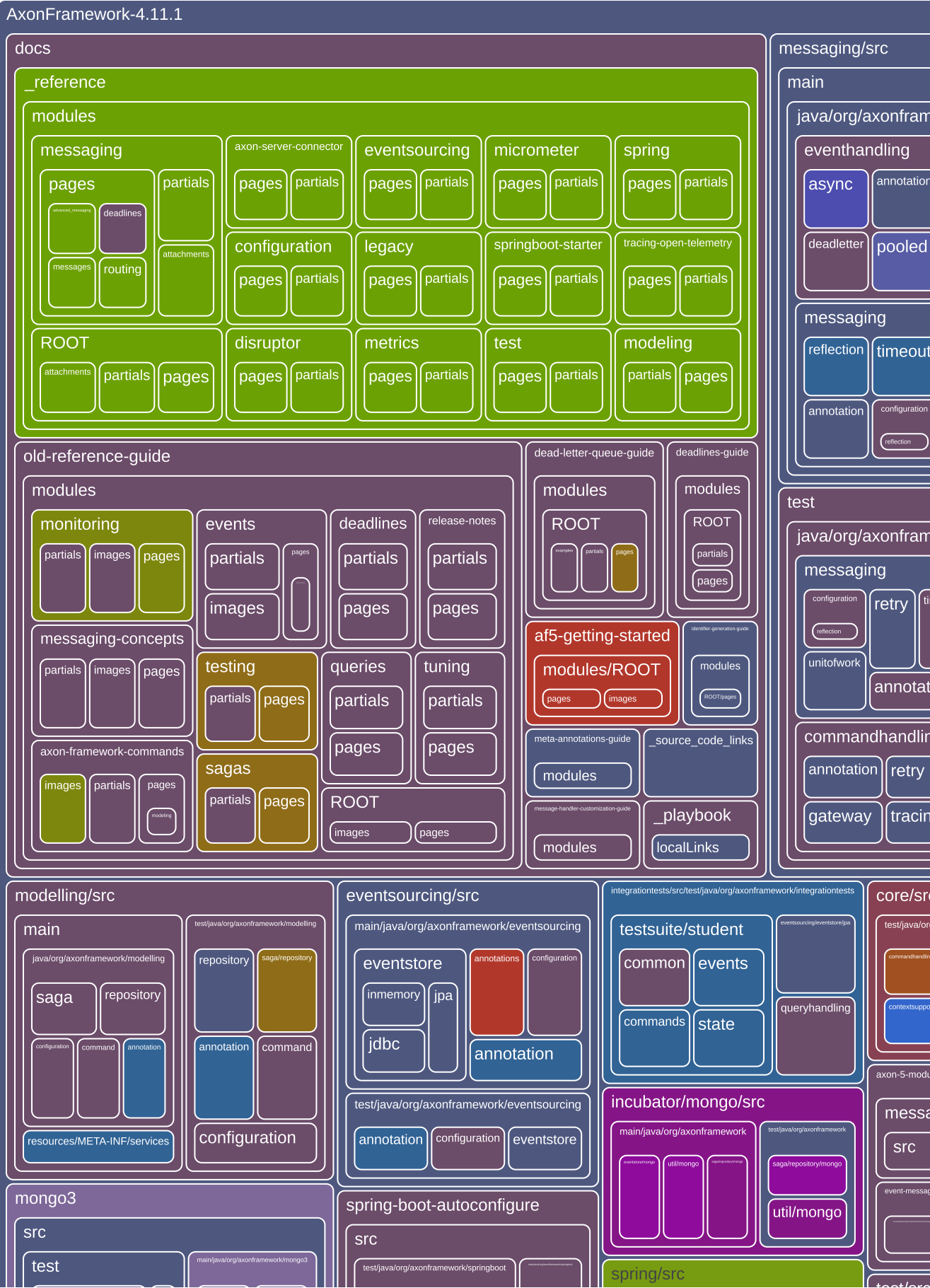
AxonFramework-4.11.1





Second author per directory

Second author with the second highest number of commits

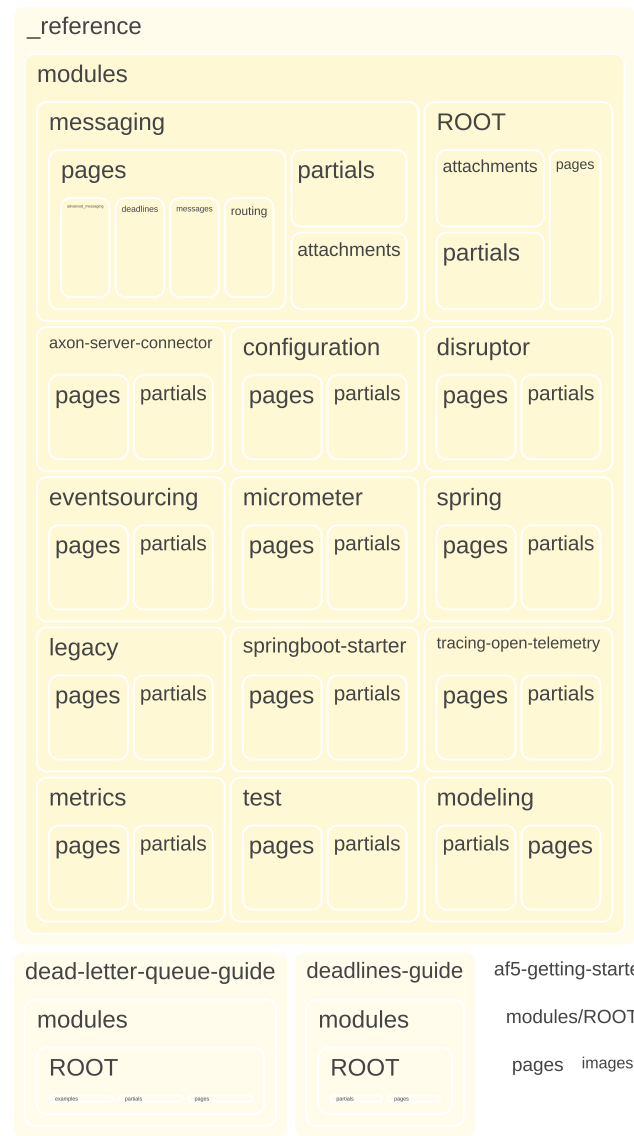




Days since last commit

AxonFramework-4.11.1

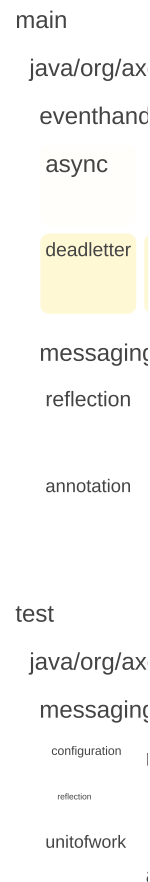
docs



old-reference-guide

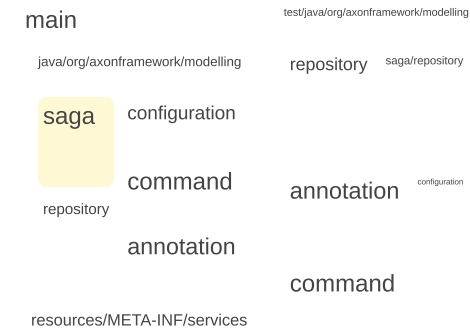


messaging/src



commandl
annotatio
gateway

modelling/src



eventsourcing/src



integrationtests/src/test/java/org/axonframework/integrationtests



spring/src



spring-boot-autoconfigure



src

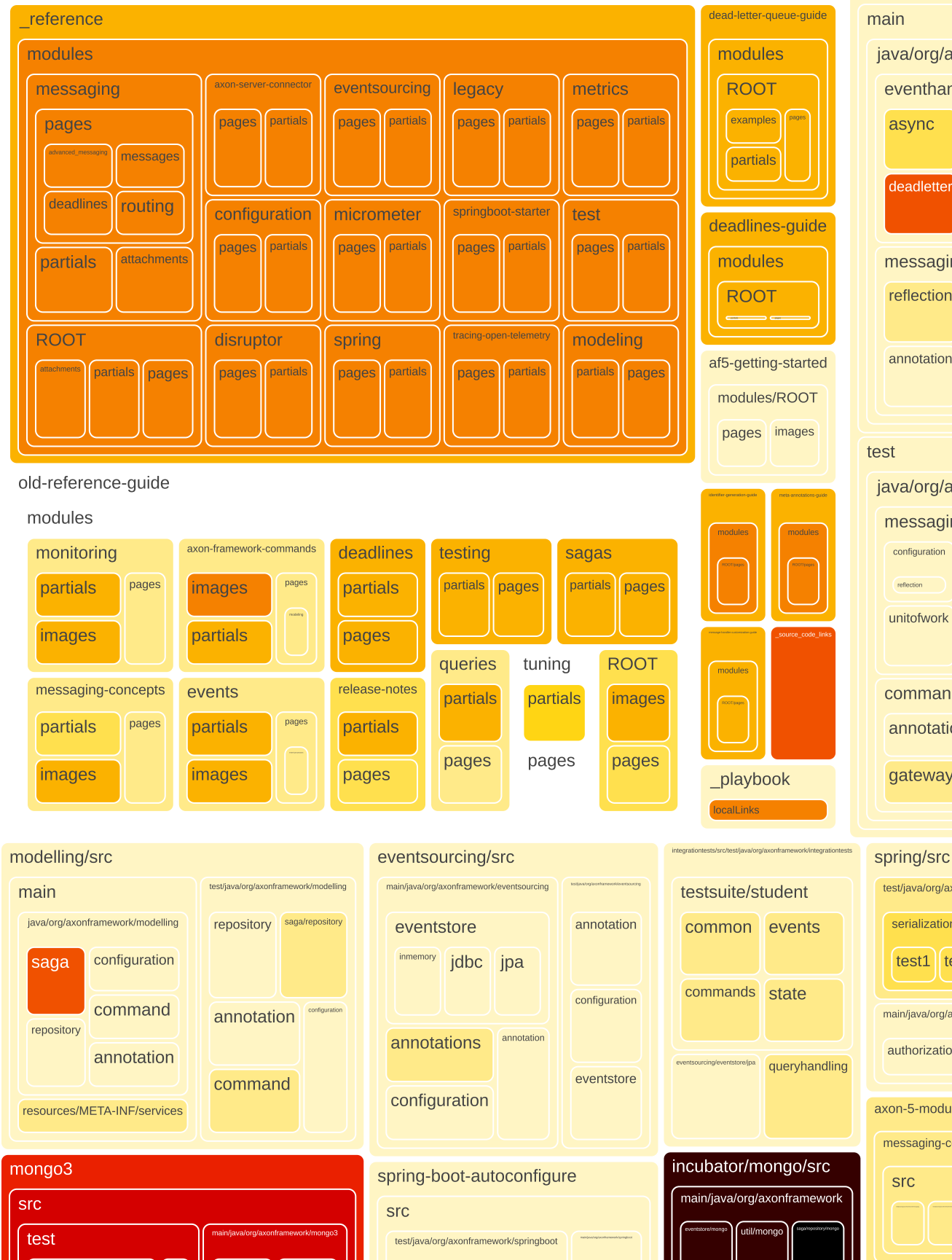


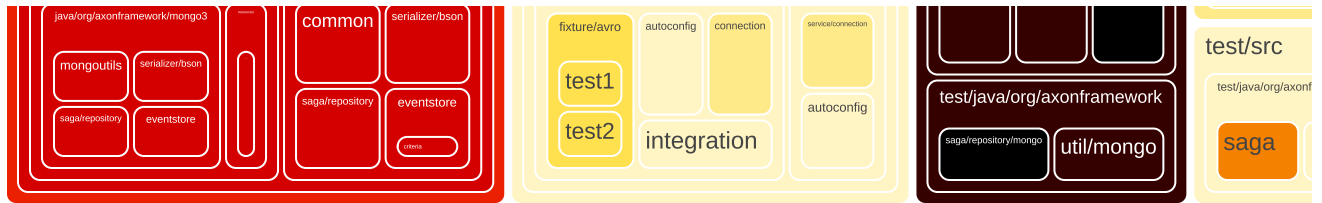
Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.11.1

docs





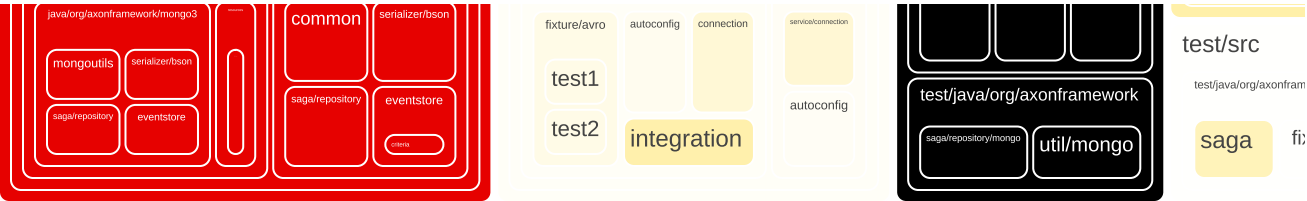
Days since last file creation per directory

Days since last file creation

AxonFramework-4.11.1

docs



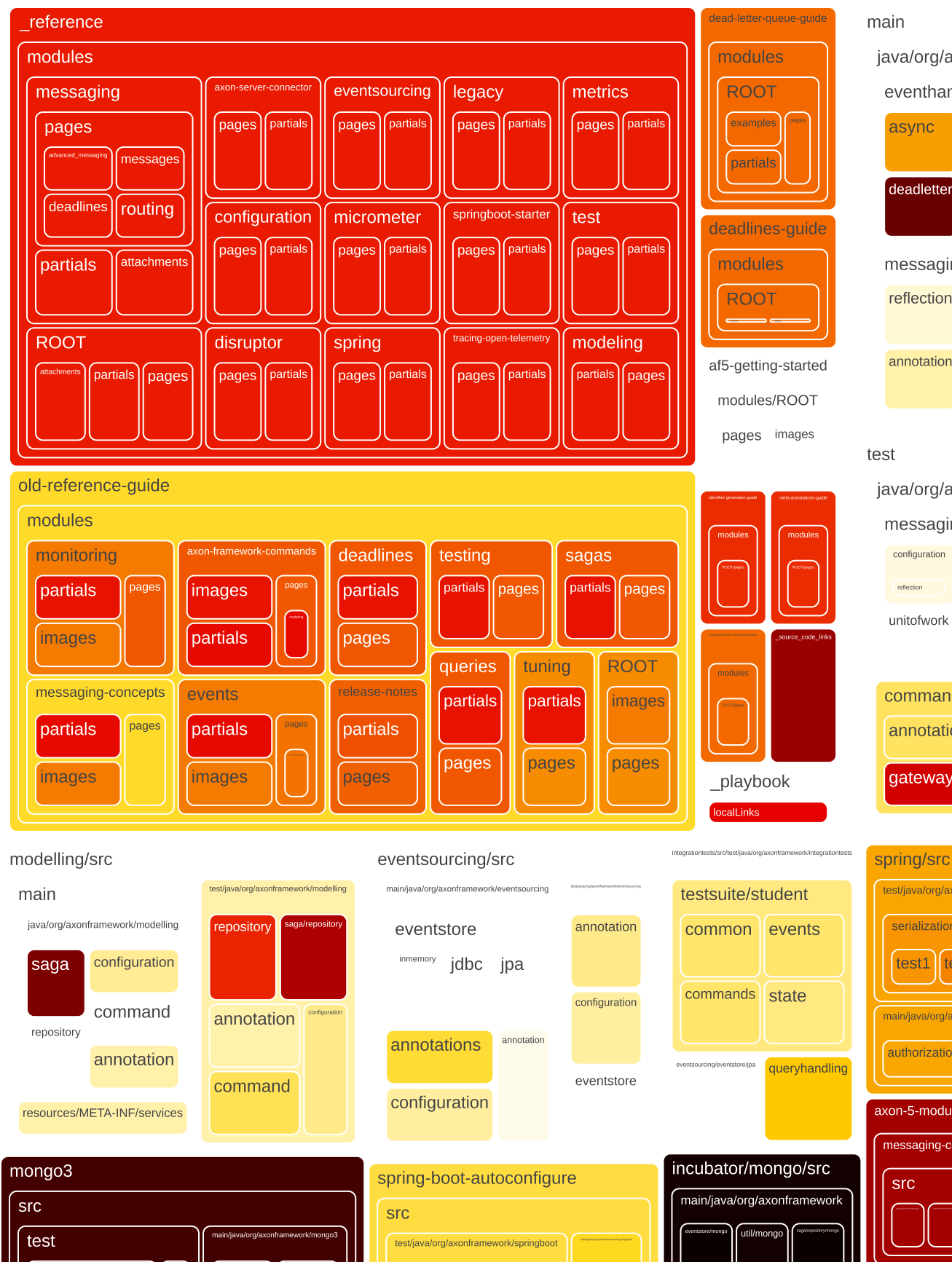


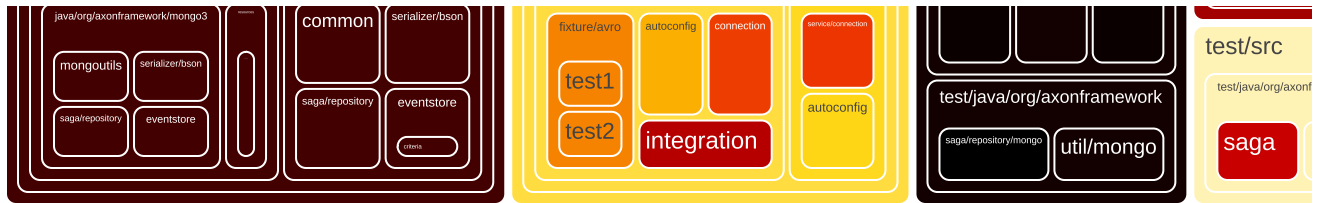
Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1

docs



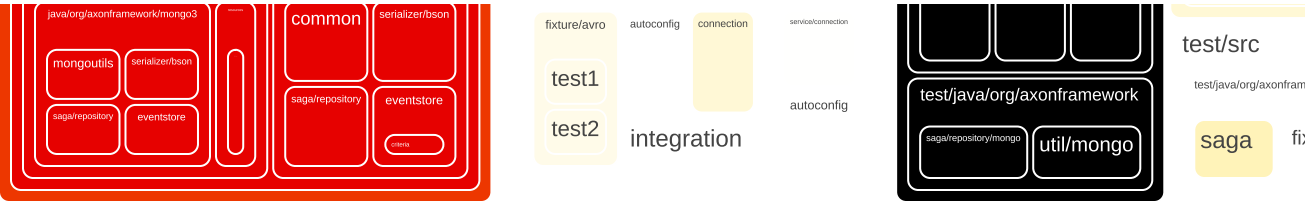


Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1



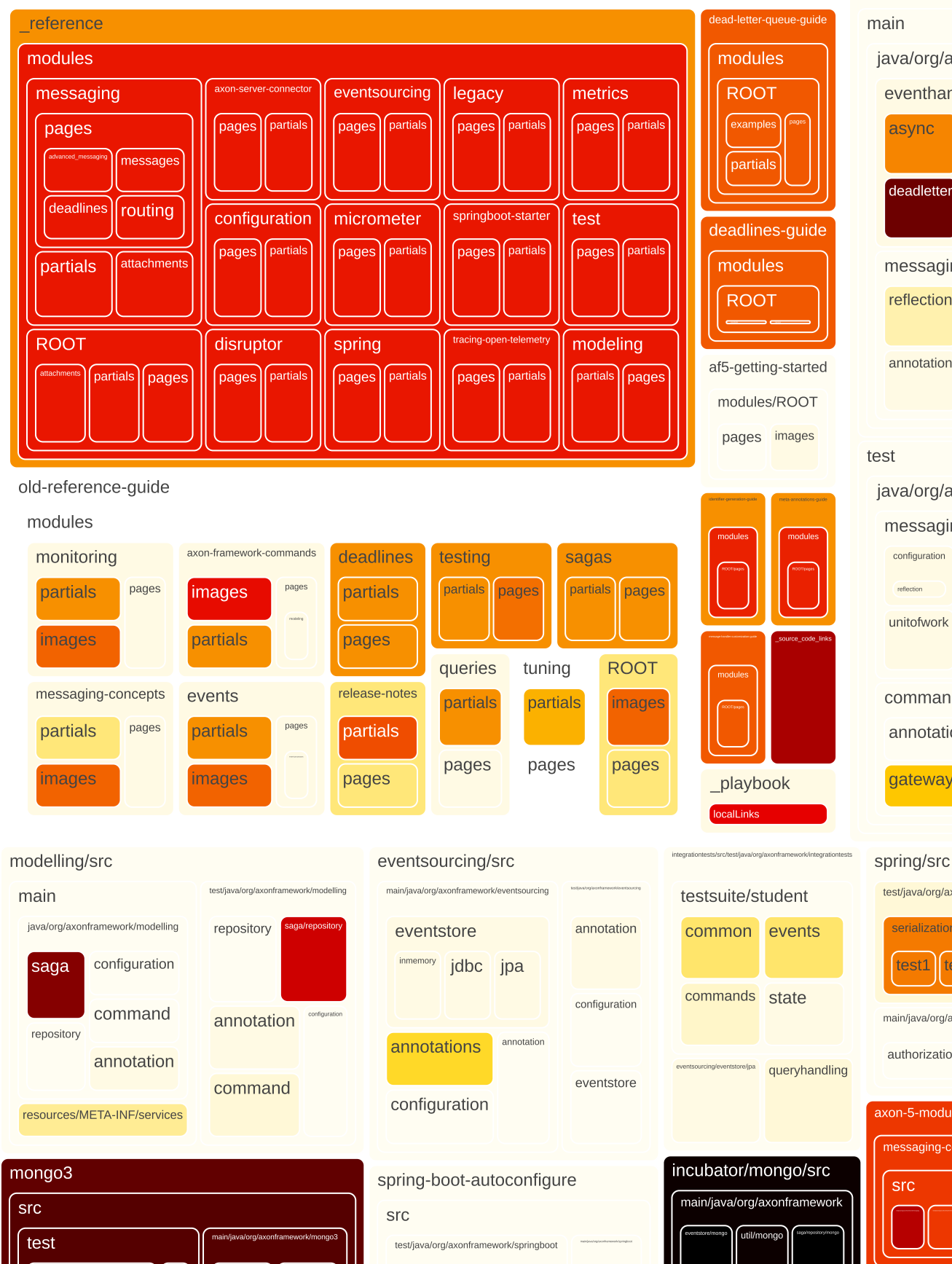


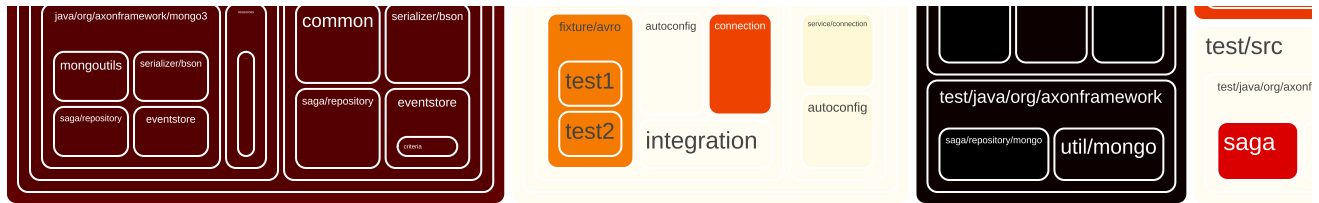
Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-4.11.1

docs

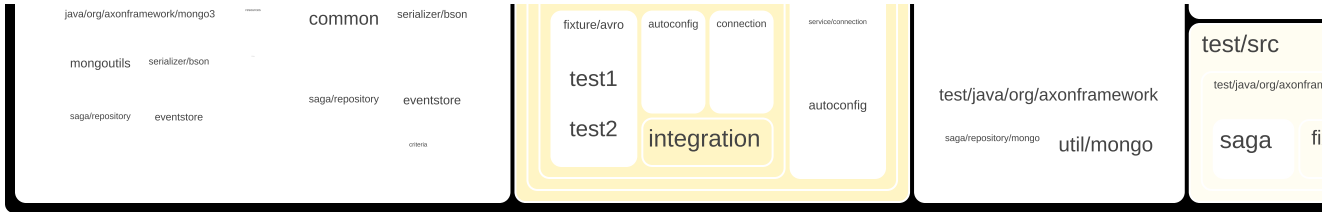




File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

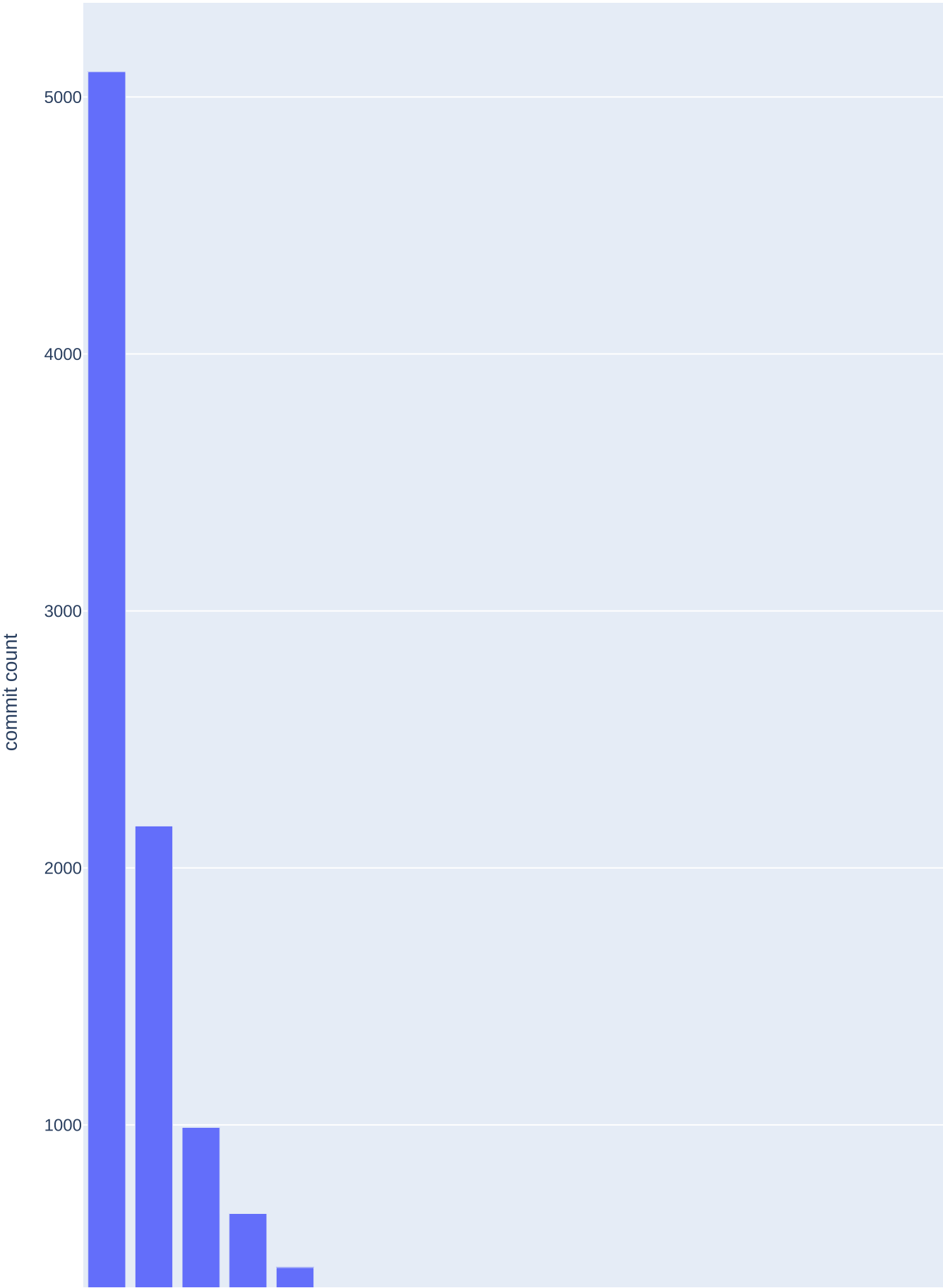
Sum of commits that changed more than 30 files (each) = 1211
Max changed files with one commit = 4041

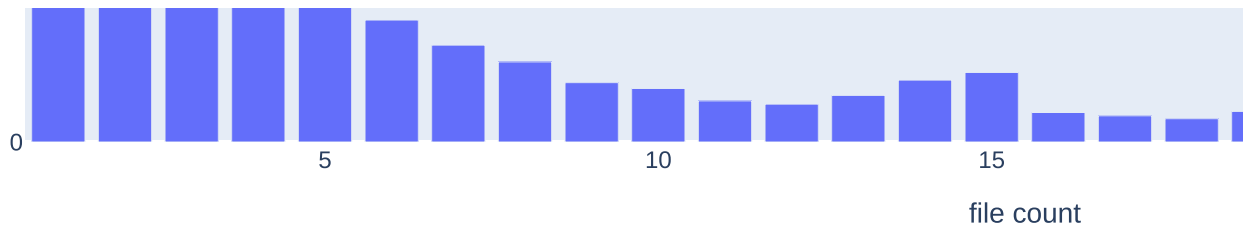
	filesPerCommit	commitCount
count	377.000000	377.000000
mean	403.740053	34.925729
std	568.423371	292.901288
min	1.000000	1.000000
25%	95.000000	1.000000
50%	203.000000	2.000000
75%	431.000000	5.000000
max	4041.000000	5098.000000

	filesPerCommit	commitCount
0	1	5098
1	2	2163
2	3	990
3	4	655
4	5	445
5	6	335
6	7	266
7	8	220
8	9	163
9	10	147
10	11	113
11	12	104
12	13	128
13	14	170
14	15	191
15	16	81
16	17	72
17	18	64
18	19	84
19	20	77
20	21	46
21	22	70
22	23	39
23	24	37
24	25	43
25	26	34
26	27	33
27	28	33
28	29	31
29	30	24

Bar chart with the number of files per commit distribution

Changed files per commit





Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.
- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
 "// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n ,secondCodeFile.fileName AS secondFileName\n ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n ,pairwiseChange.commitCount AS commitCount\n ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n ,commitCount\n ,fileDistanceAsFewestChangeDirectoryCommands\n // ,count(*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n // RETURN firstCodeFile.fileName AS firstFileName\n // ,secondCodeFile.fileName AS secondFileName\n // ,dependency.weight AS dependencyWeight\n // ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n // ,coalesce(relation.commitCount, 0) AS commitCount\n // ,coalesce(relation.weight, 0) AS dependencyWeight\n // ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n // ,commitCount\n // ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount"

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
0	1	14	0
1	3	14	0
2	5	20	0

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	3.0	3.000000	3.0
mean	3.0	16.000000	0.0
std	2.0	3.464102	0.0
min	1.0	14.000000	0.0
25%	2.0	14.000000	0.0
50%	3.0	14.000000	0.0
75%	4.0	17.000000	0.0
max	5.0	20.000000	0.0

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

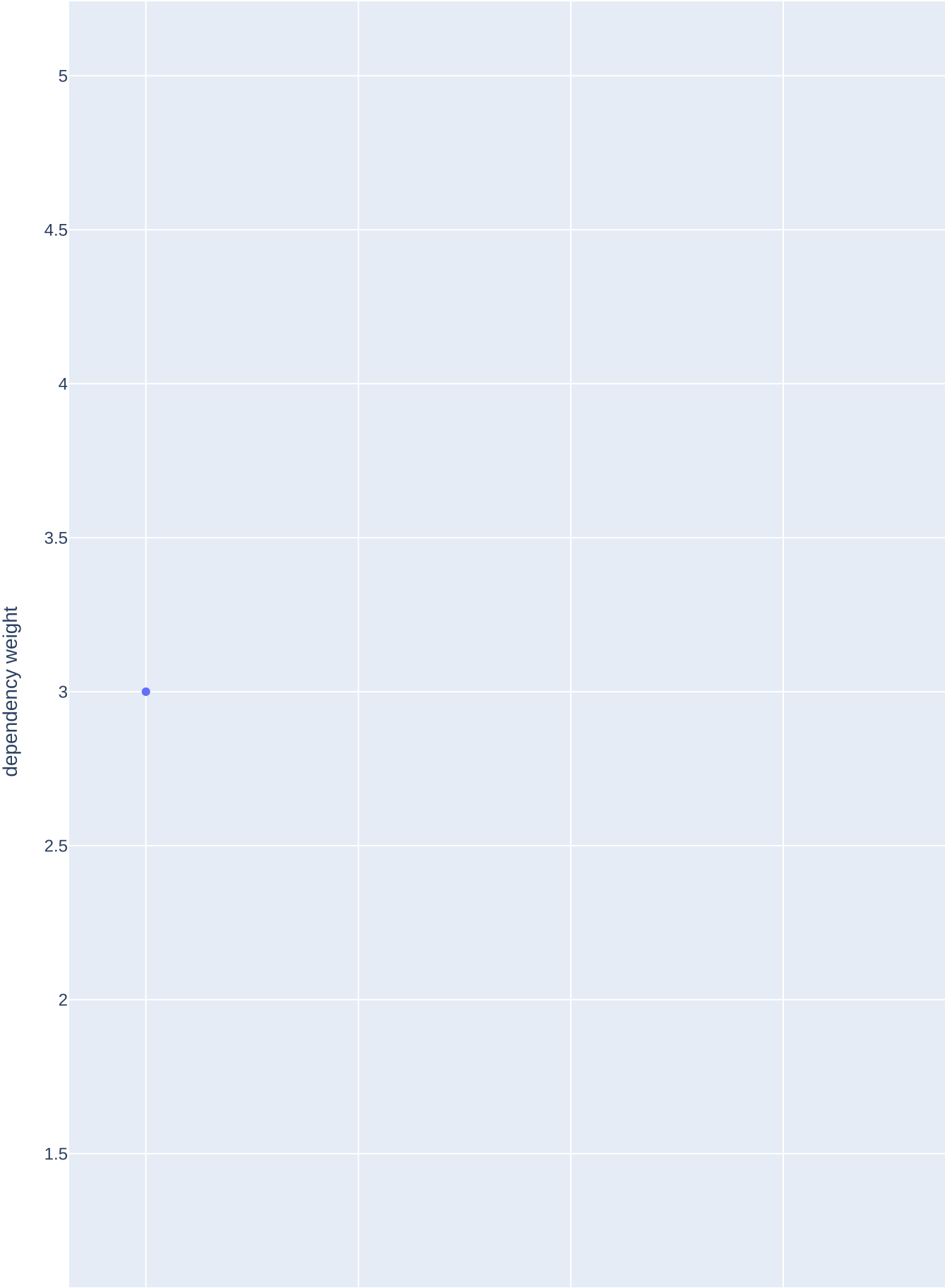
	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.866025	NaN
commitCount	0.866025	1.000000	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

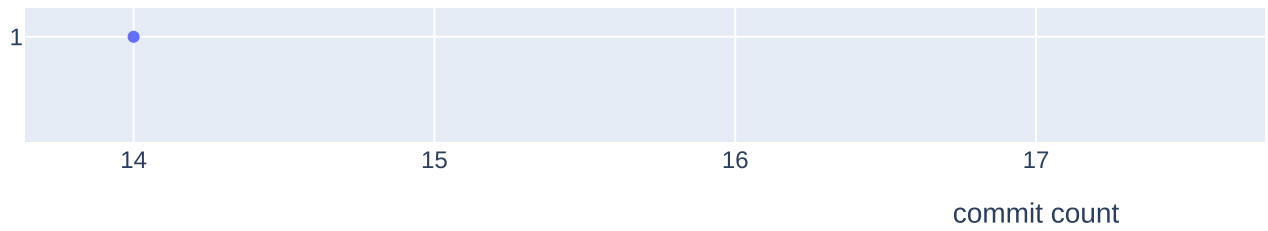
'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.866025	NaN
commitCount	0.866025	1.000000	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

Less than 5 samples are not enough to calculate p-values

Pairwise changed files: Number of changes (commitCount) vs. dependency weight





WordCloud of git authors

	word	frequency
0	Steven van Beelen	4475
1	Allard Buijze	3156
2	smcvb	822
3	Mitchell Herrijgers	476
4	Milan Savic	449
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Rene de Waele	129
9	Jetro Coenradie	123

Wordcloud of git authors

